

A two-fold increase of carbon cycle sensitivity to tropical temperature variations

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Earth system models project that the tropical land carbon sink will decrease in size in response to an increase in warming and drought during this century, probably causing a positive climate feedback^{1,2}. But available data^{3–5} are too limited at present to test the predicted changes in the tropical carbon balance in response to climate change. Long-term atmospheric carbon dioxide data provide a global record that integrates the interannual variability of the global carbon balance. Multiple lines of evidence^{6–8} demonstrate that most of this variability originates in the terrestrial biosphere. In particular, the year-to-year variations in the atmospheric carbon dioxide growth rate (CGR) are thought to be the result of fluctuations in the carbon fluxes of tropical land areas^{6,9,10}. Recently, the response of CGR to tropical climate interannual variability was used to put a constraint on the sensitivity of tropical land carbon to climate change¹⁰. Here we use the long-term CGR record from Mauna Loa and the South Pole to show that the sensitivity of CGR to tropical temperature interannual variability has increased by a factor of 1.9 ± 0.3 in the past five decades. We find that this sensitivity was greater when tropical land regions experienced drier conditions. This suggests that the sensitivity of CGR to interannual temperature variations is regulated by moisture conditions, even though the direct correlation between CGR and tropical precipitation is weak⁹. We also find that present terrestrial carbon cycle models do not capture the observed enhancement in CGR sensitivity in the past five decades. More realistic model predictions of future carbon cycle and climate feedbacks require a better understanding of the processes driving the response of tropical ecosystems to drought and warming.

Climate variability related to El Niño/Southern Oscillation has a dominant role in forcing year-to-year variation in CGR, particularly the forcing of temperature variability^{9–13} (Fig. 1a and Extended Data Fig. 1). The observed positive correlation between CGR and temperature reflects the direct impacts of temperature variations in driving variations of tropical carbon fluxes^{9,10}, rather than, in reverse, the greenhouse effect of atmospheric CO₂ (Methods). Because tropical temperatures are thought to be close to the optimal photosynthetic temperature¹⁴, rising tropical temperatures could decrease vegetation photosynthesis and increase ecosystem respiration, which amplifies the influence of temperature on ecosystem carbon exchange⁹. However, vegetation productivity and respiration both increase with higher precipitation and thus offset each other⁹. These processes lead to a weaker correlation of CGR with precipitation⁹ (with coefficient $R = -0.19$; $P > 0.10$) than that of CGR with temperature, on interannual timescale. An application of causality analysis (convergent cross-mapping¹⁵) also shows that temperature is the major climatic driver of CGR variations (Extended Data Fig. 2). Below, by analysing the atmospheric CO₂ record, we investigated how the sensitivity of tropical carbon fluxes to tropical land surface mean annual temperature (MAT) has changed over the past five decades (Methods).

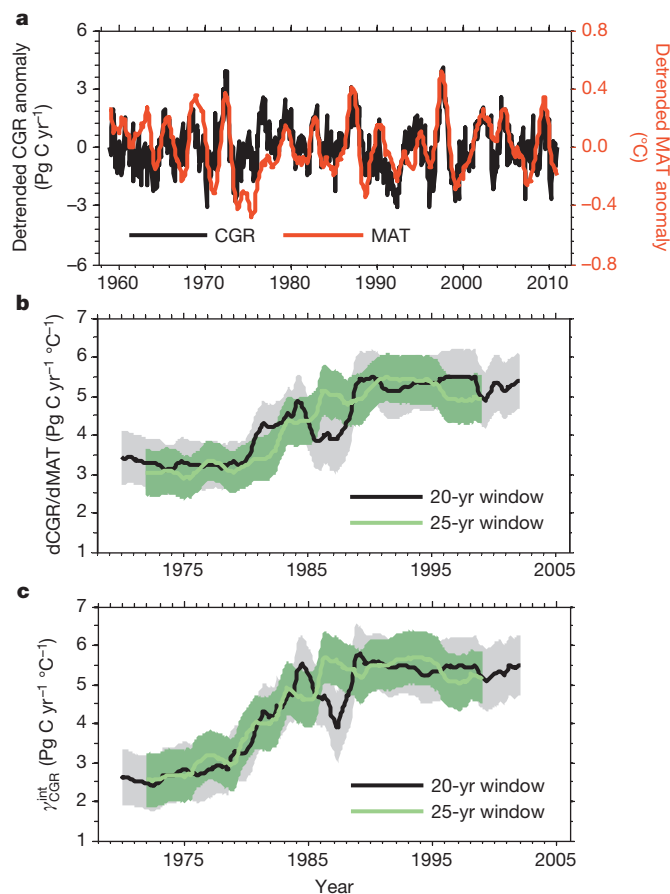


Figure 1 | Change in detrended anomalies in CGR and tropical MAT, in $dCGR/dMAT$ and in γ_{CGR}^{int} over the past five decades. **a**, Change in detrended CGR anomalies at Mauna Loa Observatory (black) and in detrended tropical MAT anomalies (red) derived from the CRU data set¹⁶. Tropical MAT is calculated as the spatial average over vegetated tropical lands ($23^\circ N$ to $23^\circ S$). The highest correlations between detrended CGR and detrended tropical MAT are obtained when no time lags are applied ($R = 0.53$, $P < 0.01$). **b**, Change in $dCGR/dMAT$ during the past five decades. **c**, Change in γ_{CGR}^{int} during the past five decades. In **b** and **c**, different colours show $dCGR/dMAT$ or γ_{CGR}^{int} estimated with moving time windows of different lengths (20 yr and 25 yr). Years on the horizontal axis indicate the central year of the moving time window used to derive $dCGR/dMAT$ or γ_{CGR}^{int} (for example, 1970 represents period 1960–1979 in the 20-yr time window). The shaded areas show the confidence interval of $dCGR/dMAT$ and γ_{CGR}^{int} , as appropriate, derived using 20-yr or 25-yr moving windows in 500 bootstrap estimates.

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The linear regression slope between CGR and tropical MAT (dCGR/dMAT) indicates the sensitivity of the tropical carbon cycle to climate interannual variability, and was used to constrain future projections of carbon cycle/climate feedbacks¹⁰. Using CGR at Mauna Loa Observatory and tropical MAT from the Climatic Research Unit (CRU) at the University of East Anglia¹⁶, we first calculated how dCGR/dMAT changes with time using a 20-yr moving window between 1960 and 2011 (both variables detrended; Fig. 1b). Within each moving window, there is a significant positive correlation between CGR and MAT ($R = 0.49\text{--}0.71$, $P < 0.01$). Furthermore, dCGR/dMAT increased from $3.4 \pm 0.4 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ (all uncertainties are s.e.; Pg C, petagrams of carbon) during 1960–1979 to $5.4 \pm 0.4 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ during 1992–2011 (Fig. 1b).

When interpreting this observed long-term increase of dCGR/dMAT, it might be argued that it does not reflect the change in the true response of CGR to temperature variations, but could be due to indirect effects of other climate variables that tend to covary with temperature, such as precipitation and solar radiation. Therefore, to isolate the role of temperature better, we defined the interannual temperature sensitivity of CGR ($\gamma_{\text{CGR}}^{\text{int}}$) as the partial derivative of CGR with respect to MAT in a multiple regression of CGR against MAT, precipitation and solar radiation (all variables detrended; Methods). The increase in dCGR/dMAT was found to reflect mainly an enhanced $\gamma_{\text{CGR}}^{\text{int}}$, rather than a change induced by covariations with the other climate variables (Extended Data Fig. 3). As shown in Fig. 1c, the value of $\gamma_{\text{CGR}}^{\text{int}}$ has increased by a factor of two, from $2.7 \pm 0.4 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ to $5.5 \pm 0.4 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ between 1960–1979 and 1992–2011. The window-by-window changes in $\gamma_{\text{CGR}}^{\text{int}}$ were fitted with three different models (the Mann–Kendall trend test, a regime shift model and a linear trend model), all of which showed that $\gamma_{\text{CGR}}^{\text{int}}$ has increased over the past five decades. However, there are too few degrees of freedom, as a result of data overlapping in neighbouring time windows. Thus, we focus below on the difference in $\gamma_{\text{CGR}}^{\text{int}}$ between the earliest time window (the first 20–25 yr) and the most recent time window (the past 20–25 yr), which are derived from fully independent subsets of the data (Methods).

We assessed the robustness of the inferred enhancement in $\gamma_{\text{CGR}}^{\text{int}}$ between the first and the last 20–25-yr period of the Mauna Loa CO₂ records. Given the significant disturbance caused by the Mount Pinatubo eruption to the global carbon cycle¹⁷, we first proposed that the cooler post-Pinatubo years characterized by enhanced carbon sinks were an unusual response of carbon sinks to MAT, which could alone explain the higher $\gamma_{\text{CGR}}^{\text{int}}$ value of the most recent two decades. Thus, we excluded the two years following the Mount Pinatubo eruption (1992–1993) from our calculation of $\gamma_{\text{CGR}}^{\text{int}}$, which had very little effect on its value, during the most recent two decades ($5.4 \pm 0.4 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$) (Extended Data Fig. 4a). To investigate whether extreme years could explain the enhancement in $\gamma_{\text{CGR}}^{\text{int}}$, we performed 500 bootstrap analyses by randomly selecting a subset of data in each 20-yr moving window to calculate $\gamma_{\text{CGR}}^{\text{int}}$. The confidence interval of these estimates is shown in Fig. 1c and confirms that the increase in $\gamma_{\text{CGR}}^{\text{int}}$ is not particularly sensitive to a few extreme years. For instance, we still obtain a significant increase in $\gamma_{\text{CGR}}^{\text{int}}$ (by a factor of 2.3–2.6) when excluding the two highest positive anomalies in CGR (Fig. 1a), which correspond to the El Niño events of 1972–1973 and 1997–1998, respectively (Extended Data Fig. 4b).

We further verified that the enhancement in $\gamma_{\text{CGR}}^{\text{int}}$ over the past five decades is robust with respect to (1) the length of the moving window for calculating $\gamma_{\text{CGR}}^{\text{int}}$ (ranging from 20 to 25 yr); (2) the application of a frequency decomposition method by singular spectrum analysis¹⁸ to separate the interannual-timescale component in CGR and MAT, instead of linear detrending (this resulted in $\gamma_{\text{CGR}}^{\text{int}}$ increasing by a factor of 1.7 for the 25-yr time windows and by a factor of 2.0 for the 20-yr time windows; Extended Data Fig. 4c); (3) the choice of climate data, using MAT from the Global Historical Climate Network¹⁹, precipitation from Global Precipitation Climatology Centre²⁰ and data on short-wave radiation²¹ ($\gamma_{\text{CGR}}^{\text{int}}$ increased by a factor of 1.9–2.1; Extended Data Fig. 4d); and (4) the choice of the other long-term CO₂ record from the South

Pole (Methods) instead of Mauna Loa ($\gamma_{\text{CGR}}^{\text{int}}$ increased by a factor of 1.7–1.9; Extended Data Fig. 4e). Overall, by analysing the histogram distribution of $\gamma_{\text{CGR}}^{\text{int}}$ values obtained with different data sources and different sensitivity tests described above (Methods), we conclude that $\gamma_{\text{CGR}}^{\text{int}}$ has robustly increased from $2.6 \pm 0.5 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ to $4.8 \pm 1.0 \text{ Pg C yr}^{-1} \text{ } ^\circ\text{C}^{-1}$ (Fig. 2a). This represents an average relative increase by a factor of 1.9 ± 0.3 (Fig. 2b). This enhancement is very unlikely to have resulted by chance ($< 10^{-4}$).

Finally, we applied the same analysis using the land sink calculated as a residual from all other terms of the global carbon budget, as an alternative to CGR for measuring variations in the global land carbon balance²², and also verified a doubling of $\gamma_{\text{Land Sink}}^{\text{int}}$ (Extended Data Fig. 4f). This test demonstrates that enhanced $\gamma_{\text{CGR}}^{\text{int}}$ is not explained by any known change in the variability of fossil fuel fluxes, ocean fluxes or emission from land-use change.

Tropical lands have experienced increasing drought in the past five decades^{23,24}, particularly from the mid 1970s to the early 1990s (Extended Data Fig. 5), when the increase in $\gamma_{\text{CGR}}^{\text{int}}$ began (Fig. 1c). On the regional scale, the more recent period (1992–2011) is drier than the earlier period (1960–1979), mainly in northern and central Africa and eastern South America (Extended Data Fig. 6), where a significant correlation between local MAT variability and CGR is observed (Extended Data Fig. 1). Large uncertainties exist in the spatiotemporal patterns of tropical drought changes, as indicated by the spread in the temporal evolution of three drought indices (the annual Palmer drought severity index²³ (PDSI),

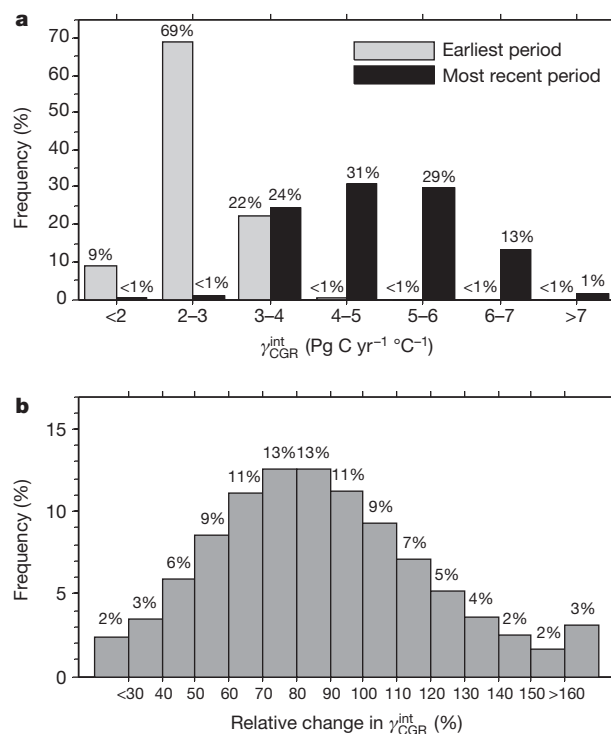


Figure 2 | Histograms of $\gamma_{\text{CGR}}^{\text{int}}$ during the earliest period and during the most recent period, and of relative change in $\gamma_{\text{CGR}}^{\text{int}}$ over the past five decades. a, Early–recent comparison; b, relative change. The relative change in $\gamma_{\text{CGR}}^{\text{int}}$ is the ratio of the calculated difference in $\gamma_{\text{CGR}}^{\text{int}}$ between the most recent period (1992–2011 for a 20-yr time-window) and the earliest period (1960–1979 for 20-yr time window) to the value of $\gamma_{\text{CGR}}^{\text{int}}$ during the earliest period. The histograms of $\gamma_{\text{CGR}}^{\text{int}}$ and of relative changes in $\gamma_{\text{CGR}}^{\text{int}}$ are calculated using bootstrap analyses, considering different data sources (CGR from Mauna Loa or the South Pole, MAT from CRU¹⁶ or the Global Historical Climate Network¹⁹, precipitation from CRU¹⁶ or the Global Precipitation Climatology Centre²⁰, and solar radiation from CRU–NCEP, ref. 21 and cloud fraction from CRU¹⁶ as a surrogate), different moving-window lengths (20–25 yr) to calculate the $\gamma_{\text{CGR}}^{\text{int}}$, and different methods to filter out the long-term increase in CGR with time (linear regression and singular spectrum analysis¹⁸).

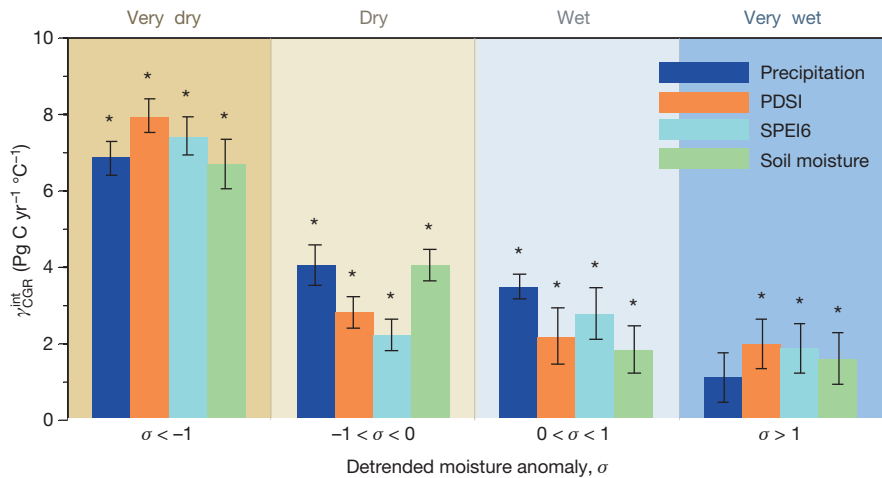


Figure 3 | $\gamma_{\text{CGR}}^{\text{int}}$ for each bin of detrended tropical moisture anomaly.

Two different drought indicators (PDSI and SPEI calculated on the 6-month scale (SPEI6)), ORCHIDEE-estimated soil moisture and precipitation were used. We grouped the 50 years' worth of data into four distinct bins of detrended tropical moisture anomaly: very dry (σ less than -1), dry (σ between

the standardized precipitation–evapotranspiration index²⁴ (SPEI) and the soil moisture estimated from the ORCHIDEE model²⁵; Extended Data Figs 5 and 6). For instance, the ORCHIDEE-estimated soil moisture shows an increase since the 1990s, whereas PDSI and SPEI are comparatively flat during the same period (Extended Data Fig. 5). To gain insight into the possible impact of tropical droughts on $\gamma_{\text{CGR}}^{\text{int}}$, we grouped $\gamma_{\text{CGR}}^{\text{int}}$ into different bins of detrended tropical moisture anomaly (Methods). Figure 3 shows that $\gamma_{\text{CGR}}^{\text{int}}$ is higher when tropical regions are drier. A Monte Carlo null experiment, assuming no change in moisture conditions, indicates that this relationship between $\gamma_{\text{CGR}}^{\text{int}}$ and drought is very unlikely to occur purely by chance ($P < 0.01$; Extended Data Fig. 7). This result suggests that soil moisture may control the response of tropical terrestrial carbon fluxes to temperature fluctuations. Many tropical ecosystems are not only subject to high temperature close to the optimal temperature of photosynthesis¹⁴, but are also subject to water limitations, at least during a dry season. Below a critical threshold for plant root water uptake, soil moisture will limit any temperature-driven increase in evapotranspiration during a warmer year. Photosynthesis will thus drop during a warmer year when soil moisture is limiting during a dry period. In turn, a reduction in photosynthesis will probably result in a larger anomaly of CO_2 input to the atmosphere²⁶, because respiration usually decreases less than photosynthesis during drought. A soil-moisture-threshold-like response of photosynthesis to temperature interannual anomalies thus can qualitatively explain why $\gamma_{\text{CGR}}^{\text{int}}$ is higher during drier periods, but additional investigation is needed to confirm that this is the main mechanism. Droughts in mid latitudes (23° – 48° north) have also increased, mostly over the past decade²⁷ (Extended Data Fig. 5), but this is inconsistent with the observed change in $\gamma_{\text{CGR}}^{\text{int}}$ (Fig. 1c). Further statistical analyses suggest that changes in mid-latitude climate must have only minor impacts on the observed $\gamma_{\text{CGR}}^{\text{int}}$ change (Extended Data Fig. 8).

Next, we explored whether terrestrial carbon cycle models can capture the observed increase in $\gamma_{\text{CGR}}^{\text{int}}$ over the past 50 years. To address this issue, we calculated the temperature sensitivity of simulated tropical net biome productivity ($\gamma_{\text{mod NBP}}^{\text{int}}$) estimated by an ensemble of five terrestrial biosphere models (Methods) previously used in ref. 22 to reconstruct historical changes in the carbon budget. We found that only one model produced an enhancement in $\gamma_{\text{mod NBP}}^{\text{int}}$ over the past five decades, and that enhancement was much smaller in magnitude than the observed change (Extended Data Fig. 9). This implies that most of the models used do not correctly capture the response of tropical carbon fluxes to climate variability⁹, probably because of biases in the response of productivity or ecosystem respiration to climate, or because

-1 and 0), wet (σ between 0 and 1) and very wet (σ greater than 1) (Methods). We then calculated $\gamma_{\text{CGR}}^{\text{int}}$ for each bin of detrended moisture anomaly, using Mauna Loa CO_2 records. The error bars indicate the s.e. of $\gamma_{\text{CGR}}^{\text{int}}$ calculated using bootstrap method (Methods). * $P < 0.05$ for $\gamma_{\text{CGR}}^{\text{int}}$ being significantly different from zero.

of inaccurate representations of soil temperature and moisture in these models. Thus, the problems present models have in reproducing the observed response of the carbon cycle to climate variability on inter-annual timescales may call into question their ability to predict the future evolution of the carbon cycle and its feedbacks to climate. Yet it should also be noted that a realistic simulation of the interannual carbon cycle dynamics provides a necessary but not sufficient test of a model's performance for the next century, because, on longer timescales, additional processes such as forest dynamics and changes in soil carbon stocks with low turnover rates become important as well²⁸.

The terrestrial carbon cycle has experienced significant changes over the past five decades^{17,29,30}. In addition to the enhanced magnitude of the land carbon sink^{17,29}, the observed significant enhancement in $\gamma_{\text{CGR}}^{\text{int}}$ found in this study provides a new perspective on a possible shift in the terrestrial carbon cycle. Increased tropical droughts are found to be related to $\gamma_{\text{CGR}}^{\text{int}}$ enhancement, highlighting the potentially important role of moisture–temperature interaction in regulating the terrestrial carbon cycle. But it should also be noted that our understanding of the mechanism of $\gamma_{\text{CGR}}^{\text{int}}$ enhancement is still limited. Further studies are needed on the response of fundamental carbon cycle processes in the tropical ecosystems and of the sensitivity of ecosystem composition and structure to environmental change. New insights could be obtained from integrated studies, combining dynamic vegetation models with data from long-term manipulative field experiments. Furthermore, new observations of atmospheric CO_2 over tropical land masses from aircraft, towers and satellites may allow for more accurate correlation with climate variables than using Mauna Loa or South Pole CO_2 data.

METHODS SUMMARY

To estimate the interannual sensitivity of CGR to tropical MAT ($\gamma_{\text{CGR}}^{\text{int}}$), we use atmospheric CGR from the two longest atmospheric records at Mauna Loa and South Pole³¹, gridded MAT from the CRU¹⁶ and from Global Historical Climate Network data¹⁹, annual precipitation from CRU and from the Global Precipitation Climatology Centre²⁰, solar radiation data from CRU–NCEP and ref. 21, and cloud fraction from CRU over vegetated areas (Methods) in the tropics. The value of $\gamma_{\text{CGR}}^{\text{int}}$ was calculated as the regression coefficient of temperature in a multilinear regression of CGR variations against variations in temperature, precipitation and solar radiation over a running time window (ranging from 20 to 25 yr). Several statistical tests (Methods) were done to assess the robustness of the increase in $\gamma_{\text{CGR}}^{\text{int}}$ over the past five decades. Two different drought indicators (PDSI²³ and SPEI²⁴) and soil moisture estimated using the ORCHIDEE model²⁵ were used to investigate the effects of drought on $\gamma_{\text{CGR}}^{\text{int}}$. We grouped the 50 years of data into four distinct bins of detrended tropical moisture anomaly (standardized departure (σ) less than -1 , between -1 and 0 , between 0 and 1 , and more than 1). For each bin of detrended

moisture anomaly, we calculated $\gamma_{\text{CGR}}^{\text{int}}$. Finally, the outputs of five terrestrial carbon cycle models²² were analysed to test whether these models could capture the observed increase in $\gamma_{\text{CGR}}^{\text{int}}$ during the past 50 years.

Online Content Any additional Methods, Extended Data display items and Source Data are available in the online version of the paper; references unique to these sections appear only in the online paper.

Received 26 July; accepted 27 November 2013.

Published online 26 January; corrected online 12 February 2014 (see full-text HTML version for details).

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Acknowledgements We thank S. Seneviratne and G. Zhang for comments and J. Gash for English editing. This study was supported by the National Natural Science Foundation of China (41125004), the National Basic Research Program of China (grant nos 2010CB950601 and 2013CB956303) and the National Youth Top-notch Talent Support Program in China. R.B.M. was funded by the NASA Earth Science Division.

Author Contributions S. Piao, P. Ciais and X.W. designed the research; X.W. performed the analysis; S. Piao, P. Ciais and X.W. drafted the paper; and P.F., R.B.M., P. Cox, M.H., J.M., S. Peng, T.W., H.Y. and A.C. contributed to the interpretation of the results and to the text.

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METHODS

Atmospheric CO₂ concentration. The measurements of monthly mean atmospheric CO₂ concentration at the Mauna Loa Observatory were obtained from the US National Oceanic and Atmospheric Administration for the period 1959 to 2011 (<http://www.esrl.noaa.gov/gmd/ccgg/trends/>) and that from the South Pole station were obtained from the Scripps Institution of Oceanography (<http://scrippsco2.ucsd.edu/>)²¹. Note that about 15% of monthly CO₂ concentrations in the South Pole station during 1959–2011, particularly during the 1960s, were gap-filled (see <http://scrippsco2.ucsd.edu> for more processing details).

Gridded climate fields. Gridded climate fields used in this study include mean annual temperature (MAT), precipitation, solar radiation, PDSI and SPEI.

Gridded MAT, annual precipitation data and cloud fraction data are from Climatic Research Unit, University of East Anglia¹⁶ (CRU TS 3.2). CRU TS 3.2 provides monthly air temperature, precipitation and other climate variables with spatial resolution of $0.5^\circ \times 0.5^\circ$ over the global land surface during the period 1901–2011. Because solar radiation is not included in the CRU TS 3.2 data set, we used cloud fraction as a surrogate for variations in solar radiation. We further obtained solar radiation from the CRU–NCEP data product (<http://dods.extra.cea.fr/store/p529viov/cruncep/>). CRU–NCEP is a gridded climate data product combining CRU TS 3.1 and NCEP–NCAR reanalysis. It provides 6-hourly meteorological forcing fields, including solar radiation, for carbon cycle model simulations (see, for example, ref. 2). Given that the discontinuity of the assimilated observations possibly causes the spurious trends in the reanalysis data set³², the monthly mean value of incoming solar radiation from the NCEP–NCAR reanalysis was corrected in CRU–NCEP to match the empirically derived monthly solar radiation using an approach based on latitude and sunshine hours³³, which was estimated to be proportional to observed cloud fraction in CRU TS 3.1. The preparation of the CRU–NCEP data set was performed in a carbon cycle model intercomparison project². Among all the climate stations used in generating this data set, about 500 are located between 23° north and 23° south, but their spatiotemporal distribution is uneven¹⁶. For example, ground measurements are particularly sparse over uninhabited area, including the Amazon basin and the tropical rainforest area in Southeast Asia. Thus, we have also used other climate data streams, including different interpolation approaches, climate reanalysis and a combination of ground and satellite measurements, to test the robustness of the inferred increase in $\gamma_{\text{CGR}}^{\text{int}}$ to uncertainties in the climate fields. The alternative climate data sets include air temperature fields of the Global Historical Climate Network Gridded Product (GHCN v3) of the National Climatic Data Center¹⁹, which has a spatial resolution of $5^\circ \times 5^\circ$; precipitation fields from the Global Precipitation Climatology Centre²⁰ (GPCC), which have a spatial resolution of $1^\circ \times 1^\circ$; and a solar radiation data set with a spatial resolution of $1^\circ \times 1^\circ$ (ref. 21). The short-wave radiation product from ref. 21 was obtained by correcting radiation in NCEP–NCAR reanalysis product with satellite-derived SRB v3.0 radiation data³⁴. All the data streams (ground measurement interpolation, climate reanalysis, and combination of ground and satellite measurements) have advantages and disadvantages, and thus could have differences despite the general consistency^{35,36}. For example, from the late 1990s onwards the CRU data set seems to be giving higher estimates of global precipitation than does GPCC³⁷. These differences in the climate data are one of the major sources of uncertainty in $\gamma_{\text{CGR}}^{\text{int}}$ estimates (Fig. 2).

The PDSI is a widely used index of drought stress and aridity change^{27,38}. We used global PDSI data provided by University Corporation for Atmospheric Research²⁴. The PDSI data have a spatial resolution of $2.5^\circ \times 2.5^\circ$ and a temporal resolution of one month from 1850 to 2010. This PDSI data set calculates potential evapotranspiration using the Penman–Monteith equation, which gives more realistic estimates than do Thornthwaite equations^{27,37}.

The SPEI is a multiscalar drought index similar to the standardized precipitation index but involving a climatic water balance considering both precipitation and evapotranspiration²⁴. The timescale of SPEI (or the standardized precipitation index) is defined so that an n -month scale considers the water surplus or deficit accumulated over $n - 1$ previous months²⁴. Previous studies suggest that timescales ranging from 3 to 12 months are best suited to monitoring various drought types³⁹. In this study, we adopted the same 6-month scale for SPEI (SPEI6) as in ref. 40, but also performed all calculations with SPEI on timescales of 3 months (SPEI3), 9 months (SPEI9) and 12 months (SPEI12). The global SPEI data set was provided by ref. 41 with a spatial resolution of $0.5^\circ \times 0.5^\circ$ and a temporal resolution of one month from 1901 to 2011.

Carbon fluxes and soil moisture from global ecosystem models. Annual net biome productivity from 1901 to 2008 was simulated by five dynamic global vegetation models (DGVMs) (HyLand (HYL), Lund–Potsdam–Jena (LPJ), ORCHIDEE (ORC), Sheffield–DGVM (SHE) and TRIFFID (TRI)) in the comparison study of ref. 2, and land carbon sink simulations in ref. 22 (<http://dgvm.ceh.ac.uk>). Because spatially explicit output from ORCHIDEE was not available before 1972 in this data set, results from an updated ORCHIDEE simulation are used. Starting from

pre-industrial equilibrium for all ecosystem carbon pools reached after spin-up, all models were forced with CRU–NCEP climate fields and increasing atmospheric CO₂ concentration for the period 1901–2008. It is important to note that the DGVM models produce only land–atmosphere CO₂ fluxes because of changed climate and CO₂; other drivers, such as nitrogen deposition, forest demography changes and land-use change, are not considered. Four DGVM models (HYL, LPJ, SHE and TRI) have a spatial resolution of $3.75^\circ \times 2.5^\circ$, and the updated ORCHIDEE simulation has a spatial resolution of $0.5^\circ \times 0.5^\circ$.

Analyses. The annual CGR of a specific month is calculated as the difference between the CO₂ concentration in that month and in the same month the previous year. Tropical MAT, annual precipitation, annual solar radiation, annual PDSI and annual SPEI were all calculated as the spatial average over all the vegetated land area between 23° north and 23° south. The vegetated land area is defined as all grid points for which the mean annual normalized difference vegetation index during 1982–2006 was larger than 0.1. Normalized difference vegetation index data were obtained from the Global Inventory Monitoring and Modelling Studies group⁴².

Interannual variation of CGR was found to be driven by climate variations, particularly by tropical temperature, on the basis of atmospheric CO₂ observations and terrestrial carbon cycle modelling (see, for example, refs 9, 10). This finding is consistent with field observations that tropical forest growth is more closely linked to variations in temperature than to variations in precipitation and solar radiation⁴³. To improve our understanding of the climate–CGR relationship, we also applied a convergent cross-mapping method¹⁸ to detect the linkage between CGR and tropical MAT (or other climate variables).

We calculated $\gamma_{\text{CGR}}^{\text{int}}$ over a running time window (ranging from 20 to 25 yr) during the study period. Linear trends, estimated by the least-squares method, were removed from the CGR and tropical climate (MAT, annual precipitation, mean annual solar radiation) time series within each time window. Then $\gamma_{\text{CGR}}^{\text{int}}$ was calculated as the regression slope of MAT in a multilinear regression of CGR against MAT, annual precipitation and mean annual incoming solar radiation (all variables detrended). Different time-window lengths were tested, up to a maximum length of 25 yr. This avoids data overlapping between the first and the most recent time window. Considering the fact that detrended CGR and MAT anomalies have 1–2-yr autocorrelation (Extended Data Fig. 10), a minimum time-window length was chosen as 20 yr. Note that, even though $\gamma_{\text{CGR}}^{\text{int}}$ calculated with shorter time windows, such as 15 yr, could be affected by autocorrelation, our main result—that $\gamma_{\text{CGR}}^{\text{int}}$ increased by a factor of two—is robust to changes in the length of the selected time window. We also calculated $\gamma_{\text{CGR}}^{\text{int}}$ using a linear mixed model that specifically accounts for temporal autocorrelation with a first-order autocorrelation function^{44,45}. The results show that including data autocorrelation in the regression does not change the value of $\gamma_{\text{CGR}}^{\text{int}}$ (Extended Data Fig. 10).

The window-by-window change in $\gamma_{\text{CGR}}^{\text{int}}$ was fitted using three different methods: the Mann–Kendall test⁴⁶, linear regression of $\gamma_{\text{CGR}}^{\text{int}}$ against time, and a regime shift model. We noted that the data overlapping in neighbouring time windows result in too few degrees of freedom to test the statistical significance of the trend in $\gamma_{\text{CGR}}^{\text{int}}$. Thus, statistical tests were made to compare $\gamma_{\text{CGR}}^{\text{int}}$ for the earliest time window with that for the most recent time window, which are fully independent from each other.

To test whether the shift in $\gamma_{\text{CGR}}^{\text{int}}$ during the study period was an artefact of our detrending method for CGR and tropical climate time series (MAT, annual precipitation and mean annual solar radiation), we applied frequency decomposition by singular spectrum analysis (SSA; see, for example, ref. 18) to extract the inter-annual variability of CGR and climate variables. Here the monthly CGR time series was decomposed into four frequency components: a high-frequency component⁴⁷, a seasonal component, an interannual component and a long-term trend component. Thresholds separating the frequency components were set a priori and constrained by the temporal resolution and length of CGR time series. Frequency components with the variability up to 4.5 months were regarded as the high-frequency component, whereas the thresholds separating seasonal, interannual and long-term trend components were respectively set to be 20 and 90 months, which were equally spaced on the logarithm axis of the frequency domain (see ref. 18 for more detailed descriptions of SSA). The interannual component of the climate variables was obtained in the same way. Then a linear regression between interannual components of CGR and those of the climate variables was calculated to estimate $\gamma_{\text{CGR}}^{\text{int}}$.

To estimate uncertainty in $\gamma_{\text{CGR}}^{\text{int}}$, we calculated it in 500 bootstrap analyses⁴⁸, for each combination of different data sources (CGR from Mauna Loa or the South Pole, MAT from CRU or from GHCN, precipitation from CRU or from GPCC, solar radiation from CRU–NCEP and ref. 21, and cloud fraction from CRU as a surrogate for solar radiation), different moving-window lengths (20–25 yr) to calculate $\gamma_{\text{CGR}}^{\text{int}}$, and different methods for filtering the long-term increase in CGR with time (linear regression and SSA). The uncertainty of change in $\gamma_{\text{CGR}}^{\text{int}}$ was taken from the 72,000,000 sensitivity tests.

To study the linkage between change in $\gamma_{\text{CGR}}^{\text{int}}$ and tropical drought, we grouped the 50 years' worth of data into four distinct bins of detrended tropical moisture anomaly (standardized departure (σ) less than -1 , between -1 and 0 , between 0 and 1 , and greater than 1). For each bin of detrended moisture anomaly, we calculated $\gamma_{\text{CGR}}^{\text{int}}$. The detrended soil moisture (SM) index anomaly is defined as the standardized departure from average of detrended SM:

$$\sigma(i) = \frac{\text{SM}(i) - \text{Mean}(\text{SM})}{\text{Std}(\text{SM})}$$

Here $\text{SM}(i)$ is the detrended SM for the year i , and $\text{Mean}(\text{SM})$ and $\text{Std}(\text{SM})$ are the average and standard deviation of the SM during the whole study period, respectively. An alternative of binning tropical moisture anomalies into eight bins on the basis of σ (less than -1.5 , between -1.5 and -1 , between -1 and -0.5 , between -0.5 and 0 , between 0 and 0.5 , between 0.5 and 1 , between 1 and 1.5 , and greater than 1.5) did not qualitatively change the result.

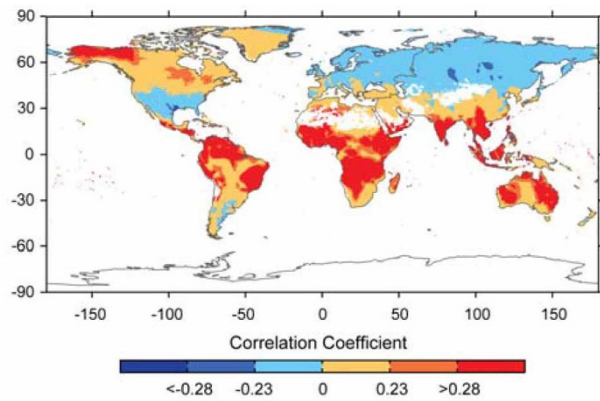
To test the robustness of the observed relationship between $\gamma_{\text{CGR}}^{\text{int}}$ and anomalies (Fig. 3), we performed a null Monte Carlo experiment. In the Monte Carlo experiment, we randomly generated artificial time series of CGR, MAT, annual precipitation and annual solar radiation following a multivariate normal distribution defined by the mean and covariations of the observed time series during 1960–2011. Because the relationship between CGR and MAT is forced to be stable during the entire period, we expected no change in $\gamma_{\text{CGR}}^{\text{int}}$ (Extended Data Fig. 7). The hypothesis to test is whether the covariations between CGR, MAT and precipitation could result in differences in $\gamma_{\text{CGR}}^{\text{int}}$ for different bins of detrended precipitation anomalies, as observed in Fig. 3.

Impacts of interannual CO_2 variations on interannual temperature variations. To estimate quantitatively the impacts of year-to-year variations in CO_2 on temperature variations, we followed refs 49, 50 to calculate the radiative forcing of CO_2 from its concentration, and used the climate sensitivity from different models. The transient temperature change due to changing CO_2 concentration is⁵⁰

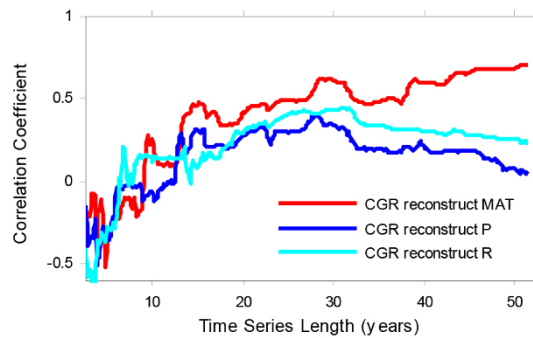
$$T_{i+1} - T_i = \frac{\alpha}{\log(2)} \log\left(\frac{C_{i+1}}{C_i}\right)$$

where T_i and C_i are the mean annual temperature and atmospheric CO_2 concentration in year i , respectively, and α is the transient temperature change in response to the doubling of CO_2 concentration. The likely range of transient climate sensitivity is $\alpha = 1.0$ – 2.5°C (ref. 51). Here we used the largest value in this range to estimate the maximum plausible impact of CO_2 greenhouse effect on temperature variations. The result suggests that the impact of CO_2 variations on interannual temperature variations is one order of magnitude smaller than the observed interannual variations in mean annual temperature. It confirms that the impact of the CO_2 greenhouse effect on temperature is not the reason for the observed significant correlation between CGR and temperature.

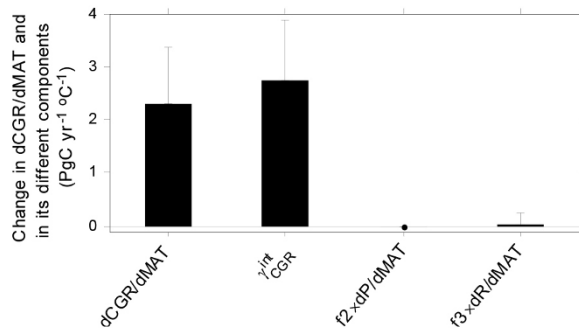
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Extended Data Figure 1 | Spatial distribution of the correlation coefficient between detrended CGR and MAT anomalies. CGR anomalies are from Mauna Loa Observatory and local MAT anomalies were derived from the CRU data set for the period 1960–2011. The correlation coefficients 0.23 and 0.28 are the critical thresholds at significance levels of 0.10 and 0.05 ($n = 52$), respectively.

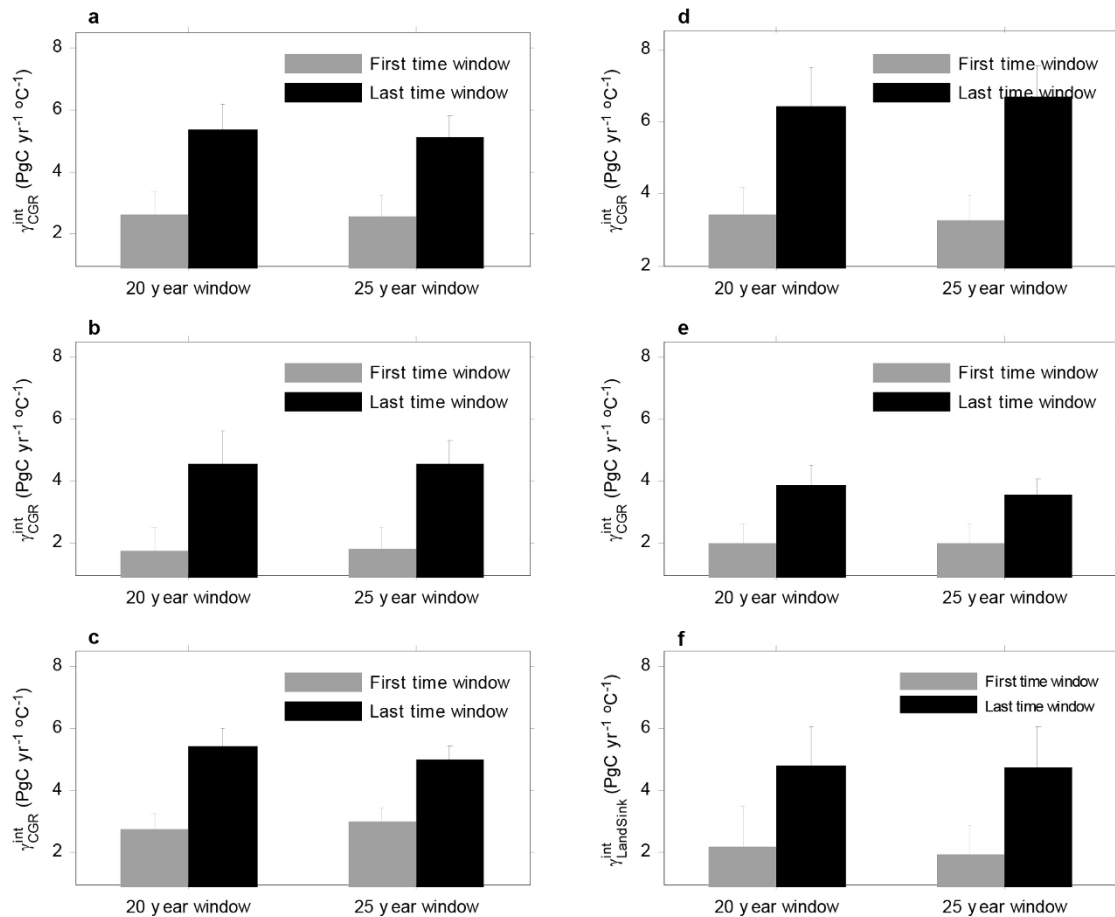


Extended Data Figure 2 | Convergent cross-mapping for reconstruction of variations in MAT, annual precipitation and mean annual solar radiation from variations in CGR. CGR data are from Mauna Loa Observatory. The CGR-reconstructed temperature curve gradually converges to a large positive correlation coefficient ($R = 0.70$), whereas the CGR-reconstructed precipitation (P) and radiation (R) curves lead to smaller correlation coefficients ($R = 0.04$ and $R = 0.23$, respectively) as time-series length increases, suggesting that CGR variations are mainly forced by temperature variations rather than by variations in precipitation and solar radiation.



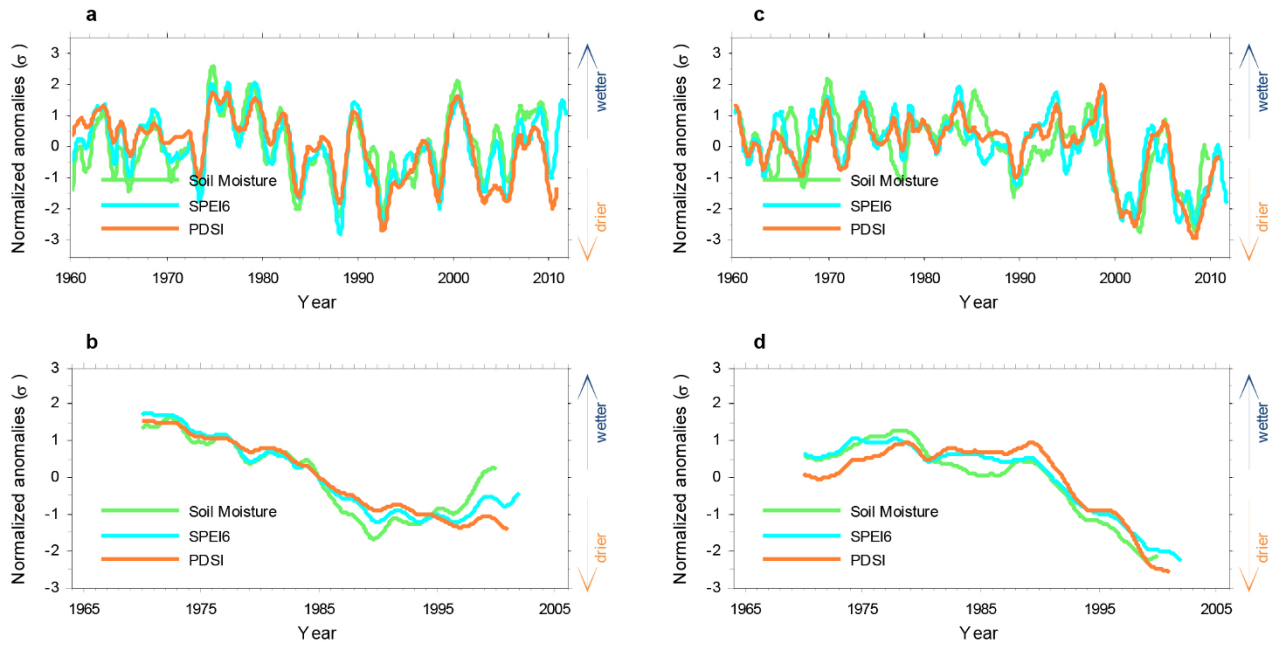
Extended Data Figure 3 | Change in dCGR/dMAT, $\gamma_{\text{CGR}}^{\text{int}}$, and the effects of interannual variations of precipitation and solar radiation on the estimate of dCGR/dMAT. The changes are calculated between the latest two decades and the earliest two decades in 1960–2011. Precipitation and radiation effects are denoted $f2 \times dP/dMAT$ and $f3 \times dR/dMAT$, respectively.

dCGR/dMAT is calculated as the slope of MAT in the regression of CGR at Mauna Loa Observatory against MAT over the tropical vegetated land. $\gamma_{\text{CGR}}^{\text{int}}$, $f2$ and $f3$ are the slopes of MAT, precipitation and radiation, respectively, in the multiple regression of CGR against MAT, precipitation and radiation over the tropical vegetated land. $dP/dMAT$ is the slope of MAT in the regression of precipitation against MAT. $dR/dMAT$ is the slope of MAT in the regression of radiation against MAT. Error bars indicate the 95% confidence interval of the corresponding value derived from 500 bootstrap estimates.



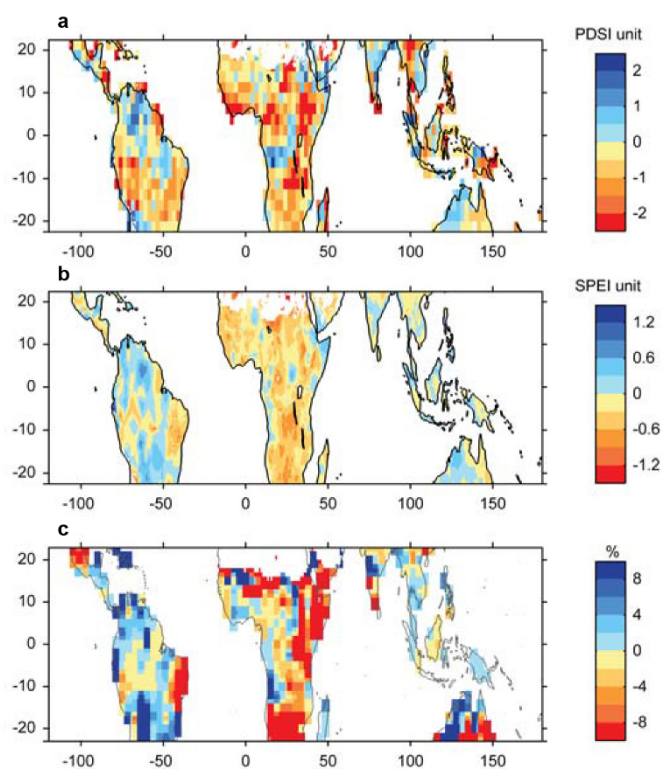
Extended Data Figure 4 | $\gamma_{\text{CGR}}^{\text{int}}$ and $\gamma_{\text{LandSink}}^{\text{int}}$ for the first and the last time window during the study period 1960–2011. For 20-yr windows, the two windows are 1960–1979 and 1992–2011. For 25-yr windows, they are 1960–1984 and 1987–2011. **a**, Data from 1992 and 1993 (post-Pinatubo years) are excluded from $\gamma_{\text{CGR}}^{\text{int}}$ estimates. **b**, Data from the record-high El Niño events of 1972–1973 and 1997–1998 are excluded from $\gamma_{\text{CGR}}^{\text{int}}$ estimates. **c**, Interannual variations of CGR and climate variables are obtained from the frequency decomposition by SSA (Methods). **d**, $\gamma_{\text{CGR}}^{\text{int}}$ is estimated with alternative climate data sets (tropical MAT is from the GHCN data set¹⁹, tropical annual

precipitation is from the GPCC²⁰ and solar radiation is from ref. 21). **e**, CGR is obtained from monthly CO₂ records at the South Pole. **f**, Interannual temperature sensitivity of the residual land carbon sink ($\gamma_{\text{LandSink}}^{\text{int}}$). The residual land carbon sink of each year is estimated from the CGR by adding the ocean sink and subtracting fossil fuel emission and emission due to land-use change²². In **a–d**, $\gamma_{\text{CGR}}^{\text{int}}$ is estimated from Mauna Loa CO₂ records. Error bars indicate the 95% confidence interval of $\gamma_{\text{CGR}}^{\text{int}}$ derived from 500 bootstrap estimates.

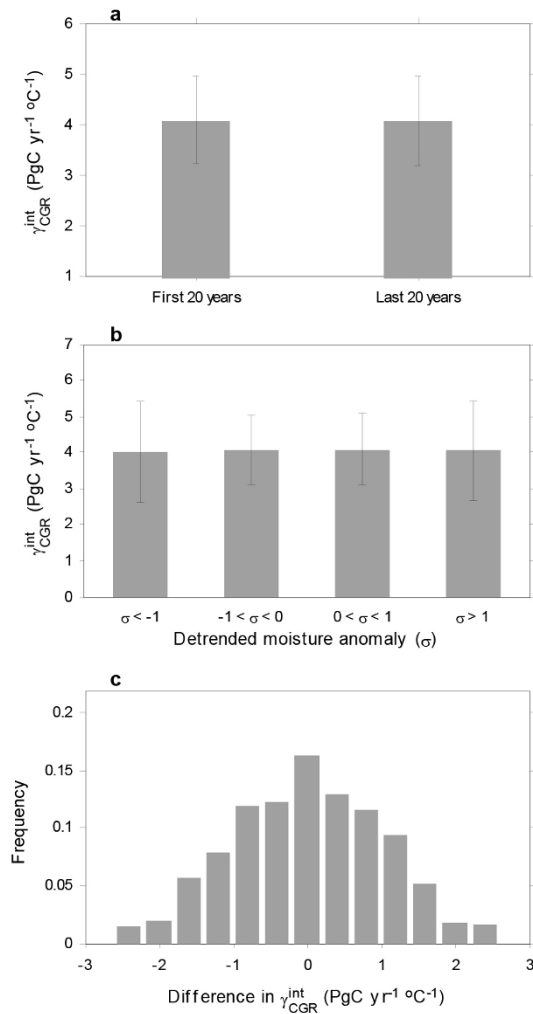


Extended Data Figure 5 | Change in drought indices and 20-yr smoothed drought indices over tropical and mid-latitude regions during the past five decades. **a**, Change in tropical (23° south to 23° north) annual PDSI, SPEI6 and ORCHIDEE-estimated soil moisture. **b**, 20-yr smoothed tropical PDSI, SPEI6 and ORCHIDEE-simulated soil moisture. **c**, Change in mid-latitude (23° north to 48° north) annual PDSI, SPEI6 and ORCHIDEE-simulated soil

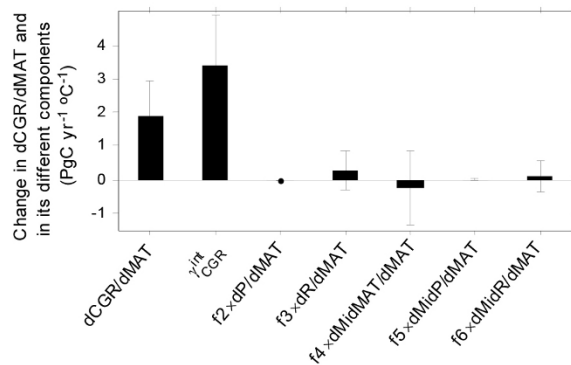
moisture. **d**, 20-yr smoothed mid-latitude PDSI, SPEI6 and ORCHIDEE-simulated soil moisture. Years on the *x*-axis indicate the central year of the 20-yr time window (for example, 1970 represents 1960–1979). All variables are normalized by their respective standard deviations. The changes in SPEI3, SPEI9 and SPEI12 are close to that in SPEI6. Note that SPEI is available till 2011, PDSI is available till 2010 and model soil moisture is available till 2009.



Extended Data Figure 6 | Spatial distribution of the difference between the latest and first 20-yr periods during the past five decades in PDSI, SPEI6 and ORCHIDEE-estimated soil moisture. a, PDSI; b, SPEI6; c, ORCHIDEE-simulated soil moisture. The changes in SPEI3, SPEI9 and SPEI12 are close to that in SPEI6.

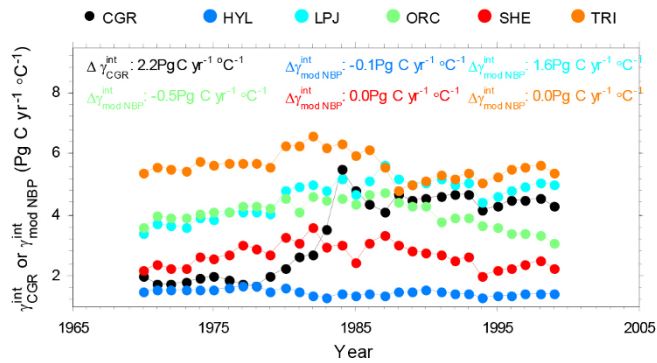


Extended Data Figure 7 | Change in γ_{CGR}^{int} and the relationship between γ_{CGR}^{int} and precipitation anomalies in the null Monte Carlo experiment. **a**, γ_{CGR}^{int} in the first and last 20 yr during 1960–2011. **b**, γ_{CGR}^{int} for each bin of detrended tropical precipitation anomalies, which are divided into four bins (standardized departure (σ) less than -1 , between -1 and 0 , between 0 and 1 , and greater than 1). Values of γ_{CGR}^{int} for different bins of detrended tropical precipitation anomaly are similar. **c**, Frequency distribution of the difference between γ_{CGR}^{int} calculated for $\sigma > 1$ (wet conditions) and γ_{CGR}^{int} calculated for $\sigma < -1$ (dry conditions). The probability of the observed difference (ranging from 5.1 to 6.0 PgC yr⁻¹ °C⁻¹; Fig. 3) occurring purely by chance is very low ($P < 0.01$). Error bars indicate the confidence intervals of the corresponding estimates.

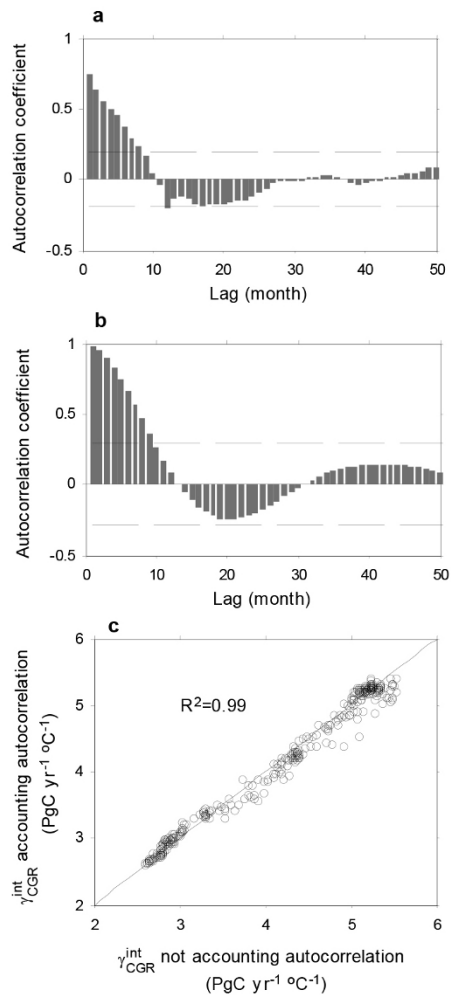


Extended Data Figure 8 | Change in dCGR/dMAT and γ_{CGR}^{int} and variational effects on estimating dCGR/dMAT between the earlier and latest two decades during 1960–2011.

The effects of interannual variations of tropical precipitation (P), tropical short-wave solar radiation (R), mid-latitude (23° north to 48° north) temperature (MidMAT), mid-latitude precipitation (MidP) and mid-latitude short-wave solar radiation (MidR) are denoted $f_2 \times dP/dMAT$, $f_3 \times dR/dMAT$, $f_4 \times dMidMAT/dMAT$, $f_5 \times dMidP/dMAT$ and $f_6 \times dMidR/dMAT$, respectively, where f_2 , f_3 , f_4 , f_5 and f_6 are respectively the slopes of P, R, MidMAT, MidP and MidR in the multiple regression of CGR against P, R, MidMAT, MidP and MidR. dy/dx represents the slope of x in the regression of y against x . Error bars indicate the 95% confidence interval of the corresponding value derived from 500 bootstrap estimates.



Extended Data Figure 9 | Change in $\gamma_{\text{CGR}}^{\text{int}}$ and $\gamma_{\text{modNBP}}^{\text{int}}$ during 1960–2008 with a 20-yr moving time window. $\gamma_{\text{modNBP}}^{\text{int}}$ is the interannual temperature sensitivity of tropical net biome productivity, estimated using five carbon cycle models (HYL, LPJ, ORC, SHE and TRI). To be consistent with model estimated annual net biome productivity, CGR of a specific year is calculated as the difference between the December Mauna Loa CO_2 concentration of the year and that of December the previous year. Positive value of $\gamma_{\text{modNBP}}^{\text{int}}$ indicates reduced anomalies of carbon sinks during warm years.



Extended Data Figure 10 | Autocorrelations in CGR and MAT, and their impacts on the estimates of γ_{CGR}^{int} . **a**, Autocorrelation coefficients for detrended anomalies of CGR from Mauna Loa during 1960–2011. **b**, Autocorrelation coefficients for detrended anomalies of MAT during 1960–2011. Dashed lines in **a** and **b** indicate 95% confidence bands. **c**, The comparison between γ_{CGR}^{int} calculated in the multiple regression of interannual variations of the Mauna Loa CGR record against interannual variations in temperature, precipitation and solar radiation (*x* axis) and γ_{CGR}^{int} calculated in the linear mixed model with same independent variables and a first-order autocorrelation function (*y* axis). Solid line indicates 1:1 ratio. This 1:1 relationship holds for γ_{CGR}^{int} derived for different time-window lengths.